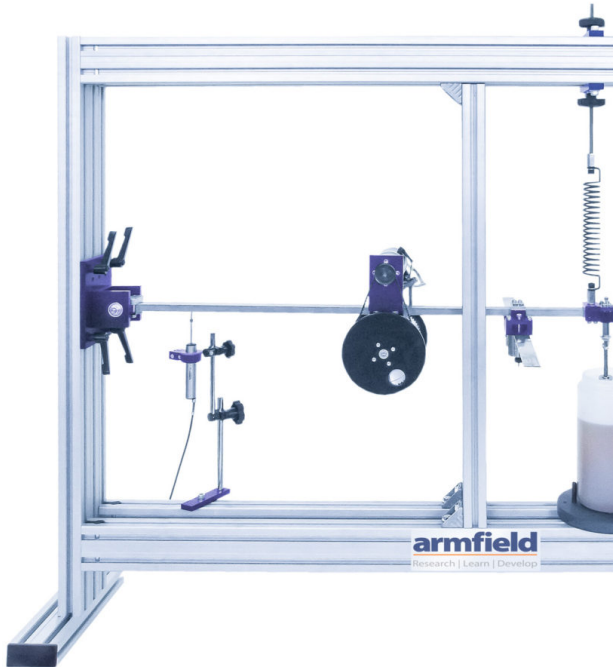




SD2 – Universal Vibration System

This system contains all the necessary parts to undertake free and forced vibration, resonance and damping.

The beam can vibrate freely by displacing the free end by a known amount, or alternatively the beam can be forced to vibrate using the supplied motor exciter. The motor exciter has integral imbalanced masses, which when rotated excites the beam to vibrate.

PRODUCT CODE: SD2

Categories: Educational, SD2 - Universal Vibration System, SV - Vibrations, SV Series - Statics & Vibrations

Tags: Calibrated Weights , Damping Discs , Excitation Frequency , Motor Exciter , SD2 , Speed Controller , Tachometer , Universal Vibration System

DESCRIPTION

This system contains all the necessary parts to undertake free and forced vibration, resonance and damping. A beam pivots at one end from a bracket and bearing attached to the Vibration Frame.

A spring is attached to the free end of the beam to enable the beam to vibrate. The horizontal position of the spring can be adjusted using the integral adjustment system, and vibration travel is restricted via factory set “stops”.

The beam can vibrate freely by displacing the free end by a known amount, or alternatively the beam can be forced to vibrate using the supplied motor exciter. The motor exciter has integral imbalanced masses, which when rotated excites the beam to vibrate. The excitation frequency of the beam is controlled using the Speed Controller and its output displayed with the Tachometer.

The excitation frequency and beam displacement are both fed into the Armfield Interface unit AIU Data Acquisition System (**supplied with the system**). Damping can be introduced into the system by using the set of damping discs, damping tank and damping media provided. The damping discs have adjustment which allows the amount of damping to be altered.

A damping absorber is also supplied that attaches to the beam, and can be adjusted to offset the resonance frequency of the system. A set of calibrated weights is supplied to vary the weight of the vibrating system.

TECHNICAL SPECIFICATIONS

Frame measures: 1380mm (L) x 1500mm (H) x 310mm (W)

Beam: 25.4mm (W) x 12.7mm (H) x 840mm (L), coated steel

Beam: 2.1kg (approx)

4 x Springs: spring rates 2.5, 0.75, 1.5, 3.0N/mm

Motor Exciter: Mass 5.1kg (approx)

8 x 5N weights

Spring steel beam: 590mm (L) x 25.4mm (W) x 1.6mm (H)

FEATURES & BENEFITS

- A comprehensive unit allowing for the study of both free and forced vibration
- Resonance and Damping
- Excellent visual demonstrator
- Vibration absorber
- Non-contact technology for beam displacement
- Motor exciter for “Forced” vibration
- Data Acquisition System
- Adjustable damping
- Four displacement springs supplied

Experimental content

- Spring stiffness
- Resonance frequency
- Active and inactive damping absorber
- Free vibration
- Forced vibration
- Damped vibration
- Damping ratio
- Tuning of damping absorber
- Amplitude response and phase response

Ordering specification

- 1 x Tachometer
- 1 x Speed controller
- 1 x Motor exciter
- 1 x Vibration absorber
- 1 x Vibrations frame
- 1 x Free & Forced vibrations module
- 4 x Springs
- 8 x 5N Weights
- Damping accessory and media
- Tools set
- Instruction manual
- Packing list
- Test sheet (transverse) vibration module

Recommended Accessories

Recommended Accessories:



SD2-1: Pendulum Module

SD2-2: Torsional Oscillation Module (Free and Damped)

SD2-3: Beam Bending (Transverse) Vibration Module

INST063: PC USB Oscilloscope

Operational conditions

- **Storage temperature:** -10°C to +70°C
- **Operating temperature range:** +10°C to +50°C
- **Operating relative humidity range:** 0 to 95%, non-condensing

Requirements

Electrical supply: 110/120V, 60Hz or 220/240V, 50Hz

PC with a USB port, running Windows 7 or above

Shipping Specifications

PACKED AND CRATED SHIPPING SPECIFICATIONS

Volume: 1.23m³

Gross: weight 116kg

Dimensions

Length: 1.580m

Width: 0.940m

Height: 1.270m

Order Codes

- **SD2:** Universal Vibration System (Includes Tachometer, Speed Controller, Motor Exciter, Vibration Absorber, Vibrations Frame and Free & Forced Vibrations Module)
- **SD2-1:** Pendulum Module
- **SD2-2:** Torsional Oscillation Module (Free and Damped)
- **SD2-3:** Beam Bending (Transverse) Vibration Module
- **AIU:** Armfield Interface Unit (Supplied)
- **INST063:** PC USB Oscilloscope (Sold separately)

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